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Clinical Sports Medicine Update

A Practical Guide for the Current Use of Biologic Therapies in Sports Medicine

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Over the past decade, there has been an increased interest in the use of biologic therapies in sports medicine. Although these technologies are in relatively early stages of development, there have been substantial increases in marketing, patient demand, and clinical utilization of biologics, including platelet-rich plasma, bone marrow aspirate concentrate, and other cell-derived therapies. Direct-to-consumer marketing of biologics has also proliferated but is largely unregulated, and clinicians must accurately convey the safety and efficacy profiles of these therapies to patients. Because most insurance companies consider biologic treatments to be experimental or investigational for orthopaedic applications given the lack of high-quality evidence to support their efficacy, patients receiving these treatments often make substantial out-of-pocket payments. With a range of treatment costs among centers offering biologics, there is a need for appropriate and sustainable pricing and reimbursement models. Clinicians utilizing biologics must also have a thorough understanding of the recently clarified Food and Drug Administration guidelines that regulate the clinical use of cell and tissue products. There is a lack of consensus on the optimal preparation, source, delivery method, and dosing of biologic therapies, which has been exacerbated by a lack of sufficient experimental detail in most published studies. Future research must better identify the biologic target of treatment, adhere to better standards of reporting, and better integrate researchers, industry, and regulatory bodies to optimize applications.

Keywords: biologics; platelet-rich plasma; PRP; bone marrow aspirate; BMAC; stem cells

Over the past decade, there has been an increased interest in the use of biologic therapies in sports medicine for conditions including cartilage injuries, osteoarthritis (OA), and tendon and ligament injuries.[15,70-72,103,117,144] Biologic therapies that are used currently are usually autologous and include platelet-rich plasma (PRP), bone marrow aspirate concentrate (BMAC), and other cell-derived therapies.[71,84] While the use of biologics for the treatment of many musculoskeletal conditions is promising, these approaches have not yet achieved sufficient evidence to warrant widespread clinical application.[§] Nonetheless, marketing, patient demand, and clinical utilization of these therapies have markedly increased over the past decade.[‖]

The US Food and Drug Administration (FDA) regulates the clinical use of cell and tissue products. Clinicians utilizing biologics must have a thorough understanding of these regulations, which were clarified in November 2017.[42] Although some biologic treatment options currently utilized outside the United States have shown promise in clinical trials, some of these therapies are not currently allowed in the United States.[42,68,79,111,117] The current allowance for use of minimally manipulated autologous blood products, such as PRP and BMAC, has resulted in a large increase in the use of these therapies in orthopaedics, particularly in sports medicine applications.[12,13,104,105] In 2015, the American Academy of Orthopaedic Surgeons held a symposium on the use of biologic treatment of orthopaedic injuries, reaching a consensus that biologic therapies should be clinically implemented with a systematic translational approach based on sound scientific evidence and improved

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collaboration among investigators, regulators, and industry.[71] A subsequent symposium was held in 2018, and a white paper detailing the summary of that meeting was recently published.[18] Despite this consensus, the relatively disorganized approach that has been used to introduce biologic treatments into clinical practice contrasts with well-established models of translational research that have been successfully used and remain the gold standard across medical disciplines.[71,101]

With an increasing focus on value-based health care outcomes, there are significant cost concerns related to biologic treatment options.[104,105] While several clinical trials and meta-analyses have been published reporting clinical outcomes of PRP and BMAC for orthopaedic applications, there have been very few studies on the cost-effectiveness of PRP and none to our knowledge on BMAC or other cell-based therapies.[118,132] While being cost-conscious, clinicians marketing and utilizing biologic therapies must do so ethically, with indications for treatment based on independent high-quality evidence—ideally, randomized controlled trials (RCTs) and meta-analyses of RCTs, consistent with accepted norms of evidence-based medicine.[121] While biologic therapies hold promise for the improvement and potential cure of many musculoskeletal conditions, these technologies are still in the relatively early stages of development.[54] Additionally, clinicians should understand the cellular and molecular mechanisms underlying these treatments as well as the current evidence to support their use. In this Current Concepts review, we aim to provide a practical guide to biologic therapies for sports medicine clinicians. Namely, we discuss current regulatory guidelines, cost considerations, the current state of direct-to-consumer marketing of biologics, factors contributing to the variability in biologic preparations, and a brief overview of underlying biological mechanisms and clinical outcomes.

REGULATORY GUIDELINES

Overview

In the United States, the FDA regulates human cells, tissue, and cellular and tissue-based products (HCT/Ps) in title 21, part 1271, of the Code of Federal Regulations (CFR).[42] The FDA characterizes stem cell therapies as HCT/Ps. Interestingly, the FDA does not regulate the use of point-of-care BMAC or autologous blood products, including PRP, which have consequently become the 2 most widely used biologic therapies in the United States.[104,105] In the United States, the clinical use of cell and tissue products is regulated by 2 federal laws: the Public Health Service Act (PHSA) and the Food and Drug Cosmetics Act.[16,42] In 2001, the FDA instituted a regulatory framework to promote the safety and efficacy of HCT/Ps, which are defined as "articles containing or consisting of human cells or tissues that are intended for implantation, transplantation, infusion, or transfer into a human recipient."[42] In 2005, the FDA implemented a 3-tiered, risk-based approach to the regulation of HCT/Ps known as the "tissue rules" under section 361 of the PHSA. Category 1 products are considered to be low risk and are not subject to HCT/P oversight.[42,71] Category 2 products include lower-risk products that meet the following criteria: (1) minimal manipulation, (2) homologous use only, (3) no combination products, and (4) no systemic effect and not dependent on the metabolic activity of living cells.[42] The terms "minimal manipulation" and "homologous use" have been a source of controversy. These terms were recently clarified by the FDA[42] and are described in detail in the following section. Allogenic HCT/Ps, which have a systemic effect or are dependent on the metabolic activity of living cells, are acceptable for use only under section 361 in a first- or second-degree blood relative.[42] Category 3 products include higher-risk products that do not meet the aforementioned criteria for category 2 products as established in section 361 and are instead regulated under section 351 of the PHSA.[42] Approval of category 3 products requires a clinical evaluation process, including an investigational new drug application with the FDA, preclinical animal trials, phased clinical studies, and premarket review by the FDA before their use in clinical practice[3,4,42] (Figure 1).

Biological products are regulated according to the current good tissue practice regulations, which ensure that HCT/Ps do not contain communicable disease agents, are not contaminated, and do not become contaminated.[38] Section 361 of the PHSA established and later clarified FDA oversight on lower-risk HCT/Ps, which do not require premarket approval and the costly, time-consuming clinical evaluation process that goes with this.[42] Although minimally manipulated PRP and BMAC are not regulated as HCT/Ps, their use must be appropriately registered by clinicians, who must also follow current good tissue practices.[38,72] The preparation systems used to create PRP and BMAC are regulated under the 510(k) pathway, in which they are considered "substantially equivalent" to a currently marketed device and therefore exempt from the traditional regulatory pathway.[32]

Since the establishment of this 3-tiered system, the FDA has created guidance documents for the use of cell therapies, which were clarified in November 2017.[42] The guidance document issued in 2017 finalized drafts previously issued by the FDA.[37] In addition to guidance documents, the FDA has enforced regulation by sending warning letters to commercial businesses and clinics that market unapproved products, and in select cases it has initiated litigation against such entities.[3,16,44] Clinicians utilizing biologic therapies should understand several terms used within the FDA guidelines.[16] First, a physician is considered a "manufacturer" of HCT/Ps if he or she is involved in any step of HCT/P recovery, processing, screening, testing, storage, labeling, packing, or distribution.[42,43] An "establishment" refers to any place that manufactures HCT/Ps, including a physician's office or clinic. To lawfully market a biological product under the PHSA, an establishment must be issued a biologics license by the FDA.[43] To obtain a license, the establishment must register with the FDA, submit a list of each HCT/P manufactured, and comply with all applicable requirements as established in 21 CFR part 1271. Any establishment that performs $\geq$1

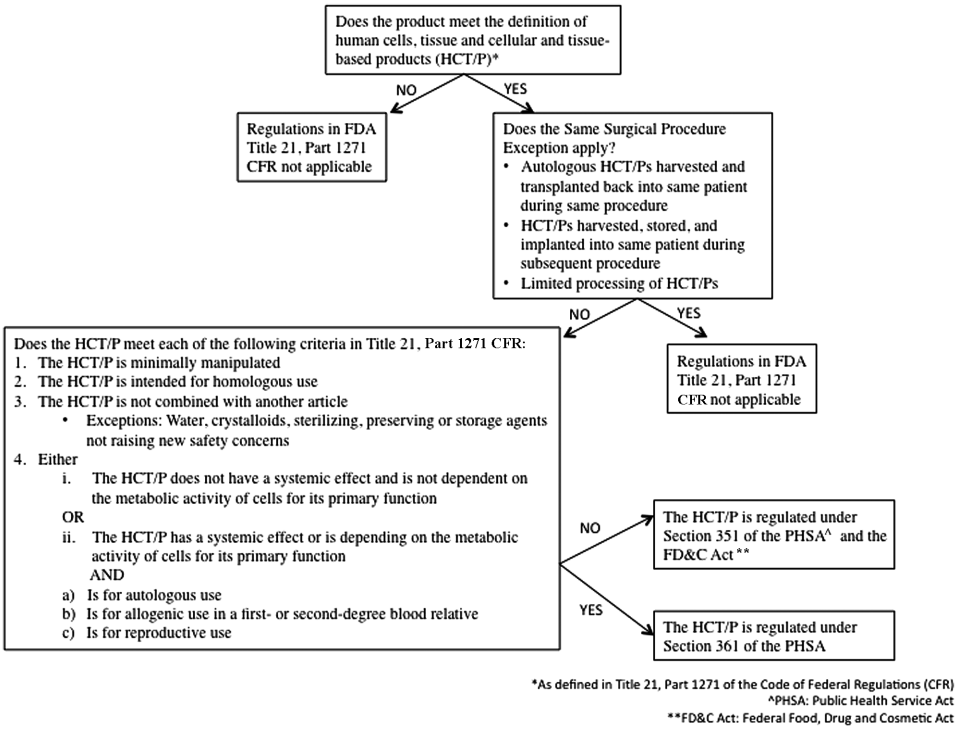

Figure 1. Overview of Food and Drug Administration (FDA) regulation of HCT/Ps.

steps in the manufacturing of HCT/Ps must register and submit a list of products with the FDA.[42]

Minimal Manipulation

The term "minimal manipulation" was recently clarified by the FDA in an effort to facilitate clinicians' and industry's understanding of how the regulatory criteria apply to HCT/Ps.[42] Part 1271 of the CFR provides 2 definitions of minimal manipulation, one of which applies to structural tissues and another to cells/nonstructural tissues.[40,42] While this discussion focuses on cells/nonstructural tissues, clinicians must be able to distinguish between cells/nonstructural tissue and structural tissues. Specifically, HCT/Ps that physically support, act as a conduit, connect, cover, or cushion are generally considered structural tissues.[42] As it relates to cells/nonstructural tissues, minimal manipulation refers to "processing that does not alter the relevant biological characteristics of cells or tissues."[42] "Relevant biological characteristics" of cells/nonstructural tissue include differentiation and activation state, proliferation potential, and metabolic activity.[42] "Processing" is defined as any activity performed on an HCT/P other than recovery, screening, storage, labeling, packing, or distribution. This includes testing for microorganisms, preparation, sterilization, preservation for storage, and removal from storage. Processing also includes "cutting, grinding, shaping, culturing, enzymatic digestion and decellularization."[42] Any processing step that alters any relevant biological characteristics is considered more than minimal manipulation. While centrifugation at the point of care to increase the concentration of stem/progenitor cells for transplantation is not generally considered more than minimal manipulation, any processing involving cell culture or ex vivo expansion is considered more than minimal manipulation because, according to the FDA, this type of processing may alter the cells' relevant biological characteristics of multipotency and capacity for self-renewal.[42]

Homologous Use

Homologous use means "the repair, reconstruction, replacement, or supplementation of a recipient's cells or tissues with an HCT/P that performs the same basic function or functions in the recipient as in the donor."[37,42] It is not necessary for the HCT/P to perform *all* of the basic functions in the recipient as in the donor to meet this definition. The basic function is what the HCT/P does or is capable of doing in its native state, as it exists in the donor.[37] HCT/Ps do not have to be used in the same anatomic location in the recipient as in the donor to perform the same basic functions, as long as it performs the same basic functions in the recipient as in the donor.[42] Examples of nonhomologous use of cells/nonstructural HCT/Ps include that from adipose tissue for the treatment of a musculoskeletal condition, such as OA or tendinopathy, because regeneration of cartilage or tendon is not a basic function of adipose tissue.[42] When applying the homologous use criteria, the FDA has discretion in determining the intended use of the HCT/Ps, as reflected by the

"labeling, advertising, and other indication of a manufacturer's objective intent."[42] If an HCT/P is intended for use as an unproven treatment for a myriad of diseases or conditions, then it is not intended for homologous use.[39,42] While most commercially available amnion-derived products appear to fit homologous use and minimal manipulation criteria, they should be marketed in a manner that conforms to these guidelines.[42,79,111] Products derived from amniotic membranes that are packaged into sheets and marketed to be used as a barrier or covering are considered minimally manipulated and compliant with section 361 of the PHSA.[42] However, amniotic membrane products marketed for wound healing, to reduce scarring, or to reduce inflammation do not fit homologous use criteria, as these are not basic functions of the amniotic membrane.[42] Additionally, processing of amniotic membrane, such as grinding or lyophilization, is considered more than minimal manipulation, as this type of processing alters its characteristics to serve as a barrier. Amniotic fluid is not considered an HCT/P and is therefore not regulated under 21 CFR part 1271.[42]

Same Surgical Procedure Exception

Under certain circumstances, the FDA does not require compliance with the requirements of 21 CFR part 1271. Namely, procedures in which autologous HCT/Ps are implanted into the same individual during the same surgical procedure are exempt under the same surgical procedure exception.[43] The FDA previously established that it would not assert any regulatory control over cells or tissues that are removed from a patient and transplanted back into that same patient during a single surgical procedure.[41,43] Furthermore, autologous tissue not used during the same procedure in which it is harvested but is instead stored to be used in a subsequent application on the same patient is also exempt under the same surgical procedure exception, as the FDA considers the 2 applications "essentially a single continuous procedure."[43] Under the same surgical procedure exception, processing must be limited to rinsing, cleansing, sizing, shaping, labeling, and storage.[43] While limited handling—including centrifugation or filtration to remove debris, such as blood or bone particles—is considered minimal manipulation under the same surgical procedure exception, cell isolation, cell expansion, cell activation, or enzymatic digestion does not qualify for this exception.[43]

REIMBURSEMENT AND INSURANCE CLIMATE

Over the past 2 decades, the use of autologous blood products (eg, PRP) and stem cell therapies (eg, BMAC) for orthopaedic conditions has markedly increased in the United States.[7,73,104,105,112] Although the clinical use of biologic therapies is becoming increasingly common for the treatment of musculoskeletal conditions, their value has not been well established.[7,13,73,100,104,105,112] Nonetheless, the use of PRP is projected to increase by 11.9% annually through 2020,[97] with a projected industry value of $350 million, with most of this projected growth attributable to orthopaedic applications.[126] Similarly, the cell therapy market has been expanding in recent years, with orthopaedic applications representing 49% of tissue engineering and stem cell industry expenditures.[63,92] Current consensus is emerging that many of these minimally manipulated preparations in current use as "stem cells" are distinct from laboratory-purified, culture-expanded cell preparations and are more appropriately labeled connective tissue progenitors.[101]

The relatively high cost of these biologic treatments for orthopaedic applications must be a consideration in discussion of these treatment options with patients.[104,105] Most major insurance companies have confirmed noncoverage policies on biologic treatments owing to a lack of high-quality evidence to support their efficacy. Most health insurance plans currently consider PRP, other autologous blood injections, and cell therapies (including BMAC) to be experimental or investigational for orthopaedic applications and cannot be billed by providers for these services. The Centers for Medicare and Medicaid Services covers only the use of autologous PRP for patients with chronic nonhealing diabetic, pressure, and /or venous wounds meeting certain criteria but does not cover the use of PRP for other orthopaedic conditions.[36] This has resulted in patients' making substantial out-of-pocket payments for biologic treatments.[6,23,104,105]

As such, clinicians utilizing or considering the use of biologics should know the actual cost to the patient. Few studies have reported on the price and cost of PRP therapy.[23,105,118,132] The price of a PRP injection in 2012 was estimated to be $200 to $500,[23] which increased to $450 to $2500 per injection in 2015.[132] A 2018 study by Piuzzi et al[105] reported a mean price for same-day unilateral knee injections of $714 (95% CI, $691-$736; range, $380-$1390) among 153 PRP clinics nationwide providing pricing information. Other studies actually reported higher estimated costs of PRP for various musculoskeletal conditions,[118,132,142] likely in part because the administration of PRP in hospitals and ambulatory surgery centers can often include significant facility fees that may not be charged in clinic settings.[129,142] Two studies assessed the cost-effectiveness of PRP after arthroscopic rotator cuff repair, with both demonstrating that at current costs, the use of PRP is not cost-effective as an augment to rotator cuff repair.[118,132]

While little has been published in the peer-reviewed literature on the costs of cell-based therapies, 1 recent study[104] reported pricing information from 65 centers in the United States that provided such information. For same-day unilateral "stem cell knee injections," the mean cost was $5156 (95% CI, $4550-$5762; range, $1150-$12,000). With a substantial range of treatment costs across centers, the perceived cost of labor appears to be substantial and questionable.[104] As these treatments result in high costs to patients that are generally paid entirely out-of-pocket, it will be critical to establish an appropriate and sustainable pricing and reimbursement model in the future.[76,110] To our knowledge, no studies

have assessed the cost-effectiveness of stem cell therapies for musculoskeletal conditions.

Direct-to-Consumer Marketing

The recent explosion in the direct-to-consumer marketing of biologic therapies has raised ethical and legal concerns. The growth of the internet and social media has provided new channels for the direct marketing of biologics to consumers worldwide.[121] Unfortunately, most of this direct-to-consumer advertising is unregulated.[125] Importantly, as outlined in the Regulatory Guidelines section, businesses advertising nonhomologous autologous, allogenic, culture-expanded mesenchymal stem cell (MSC), or xenogeneic therapies are not compliant with FDA regulations and may be exposing patients to unjustifiable risks and medicolegal consequences.[42,128]

Direct-to-consumer marketing and clinical applications of PRP for the treatment of OA are substantial and rising, with $93.4 million market revenue in 2015 and projected increases over the next decade.[23,105] Turner and Knoepfler[128] utilized a rigorous internet-based search to determine the number and geographic location of US businesses marketing stem cell treatments. There were more businesses marketing these treatments for orthopaedic or sports medicine conditions than any other medical condition.[128] While most businesses marketed autologous cell–based interventions, 20% marketed allogenic stem cell interventions not compliant with current FDA regulations.[42] Although PRP is not a stem cell product, it was sometimes falsely marketed as an autologous "stem cell treatment" from peripheral blood, perhaps as a marketing hook for potential customers.[128] In a recent study evaluating the claimed efficacy of PRP injections for knee OA, Piuzzi et al[105] reported that the mean marketed patient satisfaction rate was 76% (95% CI, 73.5%-78.3%; range, 49%-97%). In a related study[105] on stem cell treatments for knee OA, the authors found that these businesses reported a mean clinical efficacy of 82.2% (95% CI, 79.0%-85.5%; range, 55%-100%) for stem cell injections to treat knee OA. A study[108] reporting on social media content related to cell-based therapy for musculoskeletal conditions reported that 94% of social media posts had a positive tone, with 83% of all posts made by a business or commercial organization. The only negative social media posts came from patients, who had 60% positive and 40% negative posts overall. Interestingly, the authors found that approximately half of all advertising posts originated from a single business that recruited physicians to market their cell-based therapies on social media.[108]

Much of the publicity of stem cells and other biologic therapies by the nonscientific community has been optimistic, oftentimes beyond that of well-established therapies for the same conditions being treated.[141] While laws in the United States prohibit false, misleading, or deceptive advertising and require marketing claims to be supported by appropriate evidence, regulations prohibiting the use of testimonials and direct-to-consumer advertising of prescription drugs are more lax than in other countries.[133] As clinicians, we must be careful to accurately represent the safety and efficacy of biologic interventions, as misrepresentations, false, or exaggerated claims can be perpetuated.[9] As the use of PRP and BMAC is not subject to the same strict regulations as other biologic therapies as outlined here, clinicians must hold themselves accountable for the validity of their therapeutic claims when recommending and utilizing these treatments. If patients spend their limited resources on therapies that may be less effective than the established standard of care, they may forgo other treatments that may have been more effective.[121] Clinicians must remember that, even in our regulated health care market, there still frequently exists an asymmetry of information in which patients must rely on providers' knowledge.[121]

CLINICAL INDICATIONS FOR BIOLOGIC THERAPIES

Over the past decade, there has been a marked increase in the number of studies reporting outcomes of biologic therapies for orthopaedic indications.[86,88,144] We provide a brief overview of clinical indications and outcomes based on a literature search, including the most recent primary research studies and systematic reviews/meta-analyses. A September 2018 search of ClinicalTrials.gov revealed 884 clinical trials evaluating MSCs, 393 evaluating PRP, and 56 evaluating platelet-poor plasma for musculoskeletal conditions.

TENDON INJURY

Tendon healing requires the interaction of multiple cytokines, and it is likely that no single growth factor will be effective in promoting healing.[88] A number of growth factors present in PRP were shown to increase tenocyte proliferation and extracellular matrix production while protecting against oxidative stress and having anti-inflammatory effects.[1,62,66,88,91,115]

Rotator Cuff Tears

Platelet-Rich Plasma. The ClinicalTrials.gov search revealed 13 clinical trials investigating the use of PRP for treating rotator cuff tears. To date, clinical trials of PRP used in an effort to augment rotator cuff healing have reported equivocal results.

The use of PRP as an augment to rotator cuff repair appears safe, as only 2 cases of infection and no additional complications were reported in a recent meta-analysis.[109] Table 1 summarizes meta-analyses of level 1 and 2 studies investigating the use of PRP in the setting of rotator cuff repair. At this point, there remains a lack of evidence to support the use of PRP for the augmentation of rotator cuff repair.

Cell Therapy. Previous studies demonstrated decreased resident MSC numbers among patients with rotator cuff tears.[58] The ClinicalTrials.gov search revealed 11 studies investigating the use of stem cells for treating rotator

TABLE 1
Meta-analyses of Level 1 and 2 Studies Investigating PRP in the Setting of Rotator Cuff Repair[a]

Study	Studies, n (Patients, n)	Treatment	Outcomes	Heterogeneity	Conclusions
Chen, *AJSM* (2018)[15]	8 (904)	PRP or PRFM vs control	VAS pain decreased in the PRP group at 2-6.5 mo (WMD, –0.45; 95% CI, –0.75 to –0.15; $P < .01$) and at ≥1 y (WMD, –0.53; 95% CI, –0.98 to –0.09; $P = .02$).	Significant at baseline ($I^2 = 72.0\%$, $P < .01$), short-term follow-up ($I^2 = 72.5\%$, $P < .01$), long-term follow-up ($I^2 = 76.1\%$, $P < .01$), and overall ($I^2 = 75.8\%$, $P < .01$)[b]	PRP may reduce pain associated with rotator cuff injuries
Mao, *J Orthop Surg Res* (2018)[77]	8 (219)	PRF vs control	PRF with trend toward increasing retear rate (RR, 1.30; 95% CI, 0.97-1.75; $P = .082$). No between-group difference in ASES score ($P = .066$), UCLA score ($P = .230$), Constant score ($P = .481$), or adverse events ($P = .767$).	No heterogeneity across included studies ($I^2 = 0.00\%$, $P = .614$)	PRF does not improve clinical outcomes or retear rate
Hurley, *AJSM* (2019)[59]	18 (1147)	PRP or PRF vs control	PRP decreased rates of incomplete tendon healing (17.2% vs 30.5%; $P < .05$), improved Constant score at 30 d (85.6 vs 83.1; $P < .05$), improved VAS at 30 d (2.9 vs. 4.3; $P < .05$), and final follow-up (1.2 vs 1.4; $P < .05$). PRF did not improve tendon healing or Constant scores and had longer operative time.	Significant heterogeneity with tendon healing in tears >3 cm ($I^2 = 0.8$, $P < .05$)	PRP improves healing rates, pain, and functional outcome; PRF has no benefit in healing rates or functional outcomes
Fu, *Clin Rehabil* (2017)[46]	11 (638)	PRP or PRF vs control	Standard difference in means of functional scores similar between PRF/PRP and control (0.029; 95% CI, –0.132 to 0.190; $P = .725$). Standard difference in means of VAS pain scores similar (0.142; 95% CI, –0.80 to 0.364; $P = .209$).	No heterogeneity among studies ($I^2 = 0\%$, $P = .764$)	Meta-analysis does not support use of PRP or PRFM
Zhao, *Arthroscopy* (2015)[143]	8 (464)	PRP vs control	No difference in retear rate (RR, 0.94; 95% CI, 0.7-1.25; $P = .66$), Constant score (mean difference, 1.12; 95% CI, –1.38 to 3.61; $P = .38$), or UCLA score (mean difference, –0.68; 95% CI, –2.00 to 0.65; $P = .32$).	Heterogeneity acceptable for retear rate ($I^2 = 43\%$, $P = .09$), Constant score ($I^2 = 0\%$, $P = .7$), UCLA Score ($I^2 = 0\%$, $P = .52$)	Meta-analysis does not support use of PRP
Li, *Arthroscopy* (2014)[74]	7 (417)	PC vs control	No improvement in Constant pain score (SMD, 0.99; 95% CI, –0.97 to 2.95; $P = .32$), Constant pain score (SMD, –0.20; 95% CI, –4.91 to 4.51), UCLA score (SMD, –0.16; 95% CI, –1.85 to 1.54; $P = .85$), ASES score (SMD, –2.99; 95% CI –7.92 to 1.93; $P = .23$), SST score (SMD, –0.28; 95% CI, –1.55 to 1.00; $P = .67$), or retear rates (1.11; 95% CI, 0.69-1.79; $P = .67$).	Heterogeneity significant for Constant score ($I^2 = 86\%$, $P < .0001$), UCLA score ($I^2 = 75\%$, $P = .02$), ASES score ($I^2 = 56\%$, $P = .13$), and SST score ($I^2 = 90\%$, $P = .002$)	PCs do not improve retear rate or clinical outcomes
Moraes, *Cochrane Database* (2014)[85]	6 (291)	PRP vs control	Constant scores at 1 y clinically nonsignificant difference in favor of PRP (MD, 2.47; 95% CI, 0.68-4.26). Pooled Constant and L'Insalata scores with no improvement (SMD, 0.13; 95% CI, –0.24 to 0.51); 7-d and 30-d postoperative pain scores improved with PRP but no benefit at 1 y or 2 y. Retear rates lower in PRP group after 1 y (RR, 0.5; 95% CI, 0.3-1.01) but not 2 y (RR, 0.88; 95% CI, 0.59-1.32).	Significant heterogeneity in results overall ($I^2 = 63\%$, $P = .02$)	Overall and for individual conditions, including rotator cuff tear, there is insufficient evidence to support the use of PRP for musculo-skeletal soft tissue injuries

[a]The Chen et al[15] study is level 1; the remaining are level 2. ASES, American Shoulder and Elbow Surgeons; MD, mean difference; PC, platelet concentrate; PRF, platelet-rich fibrin; PRFM, platelet-rich fibrin matrix; PRP, platelet-rich plasma; RR, risk ratio; SMD, standard mean difference; SST, Simple Shoulder Test; UCLA, University of California, Los Angeles; VAS, visual analog scale; WMD, weighted mean difference.

[b]For all studies included in analysis on tendon and ligament healing.

cuff tears. To date, clinical studies reporting outcomes after the use of bone marrow–derived MSCs have been, in general, more positive than those of PRP in this setting.[26,57] While clinical studies assessing the use of MSCs in rotator cuff repair are promising, randomized studies are lacking at this point.

Tendinopathy

Platelet-Rich Plasma. The ClinicalTrials.gov search revealed 5 studies investigating the use of stem cells and 29 investigating the use of PRP for the treatment of tendinopathy. While there is little evidence to support the use of PRP for medial epicondylitis, several RCTs and meta-analyses reported on the use of PRP in lateral epicondylitis.[60] Inconsistent results among studies may be attributable, in part, to varying concentrations of platelets, leukocytes, and other growth factors. As such, further study is necessary to determine the optimal PRP preparation to augment tendon healing.[71,72,82] While PRP may have a role in the treatment of chronic tendinopathy, no consensus has been reached in the literature at this point, with the strongest evidence in support of the use of leukocyte-rich PRP (LR-PRP).[33,72,81,83]

Cell Therapy. There have been few studies reporting outcomes of MSCs or BMAC in the treatment of tendinopathy. One study[95] reported that BMAC injection significantly improved clinical outcome scores at 2 years after injection among patients with patellar tendinopathy.

Anterior Cruciate Ligament Reconstruction: PRP

Early administration of PRP postoperatively after anterior cruciate ligament (ACL) reconstruction may accelerate ACL graft healing.[106] A systematic review demonstrated a 20% to 30% beneficial effect on earlier graft maturation,[131] and the use of PRP gel at the patellar tendon harvest site accelerated donor site healing and postoperative pain.[21] One systematic review[29] of 11 RCTs investigating the use of PRP in ACL reconstruction reported faster graft maturation in the PRP group in 6 studies, but only 1 study among the 11 RCTs reported improved clinical outcomes in the PRP group as compared with control.

ARTICULAR CARTILAGE

Osteoarthritis

Platelet-Rich Plasma. The ClinicalTrials.gov search revealed 78 studies investigating PRP in OA. Several RCTs demonstrated a positive effect of leukocyte-poor PRP (LP-PRP) as compared with placebo or hyaluronic acid (HA) for the treatment of OA.[11,96,119] While formulations with relatively high concentrations of platelets and leukocytes may be more beneficial for instances in which soft tissue healing is desired, a high concentration of leukocytes and an associated increase in inflammatory cells and degradative cytokines may not be optimal in the setting of OA.[2,24,55,138] At this point, it appears that there is more support for the use of LP-PRP in the setting of OA than LR-PRP, but an RCT directly comparing these two has yet to be published. A more refined analysis is required to determine how the proportion of leukocyte cell subtypes (lymphocytes, neutrophils, monocytes, etc) affects clinical outcomes. Furthermore, the ratio of platelets to leukocytes may be as important as the absolute number of leukocytes. There is also a paucity of literature regarding the severity of OA that may be effectively treated with PRP. Current guidelines from the American Academy of Orthopaedic Surgeons and the National Institute for Health and Care Excellence suggest that there is inconclusive evidence to support the use of PRP for knee OA.[60] Table 2 presents a summary of level 1 studies reporting on results of PRP in the setting of knee OA.

Cell Therapy. While a search of ClinicalTrials.gov revealed 105 studies investigating stem cells in OA, few clinical studies reported the effects of cell therapy in this setting.[10,13,64,67,120] Jo et al[64] reported improved knee function and reduced pain after intra-articular injection of high-dose autologous adipose-derived MSCs (1.0×10^8; laboratory purified and culture expanded) at 2-year follow-up. In another study,[67] patients with knee OA who underwent a single intra-articular injection of BMAC combined with adipose tissue aspirated from the abdomen reported improved pain and clinical function, as opposed to significantly poorer clinical improvements among patients with advanced OA. A systematic review on the use of BMAC for knee OA reported good to excellent patient-reported outcomes, with more pronounced improvements among patients with mild to moderate OA.[13]

Chondral Injury

Platelet-Rich Plasma. The ClinicalTrials.gov search revealed 4 studies investigating PRP in articular cartilage injuries. Few studies have reported outcomes of PRP in the treatment of articular cartilage injuries.

Cell Therapy. Similar to PRP, MSCs may be delivered by intra-articular injection or by surgical implantation by embedding MSCs into a scaffold.[31] While the ClinicalTrials.gov search revealed 39 studies investigating stem cells in articular cartilage injuries, few have reported clinical outcomes. To date, most studies that reported favorable outcomes with BMAC in the setting of chondral injuries utilized BMAC with a scaffold, while those using BMAC alone largely reported inconclusive results.[13,84] Very few studies have made comparisons with the scaffold alone. Table 3 summarizes studies reporting outcomes of cell-based therapy for articular cartilage defects of the knee.

While additional prospective, randomized, and longer-term studies will be needed to better assess the efficacy of BMAC in OA and for chondral defects, the current data appear to suggest that patients with earlier-grade OA and relatively smaller chondral defects may be more likely to benefit than those with severe OA, large chondral defects, or multiple chondral defects.[68] The data also suggest that cell therapy may be "symptom modifying" but

TABLE 2
Level 1 Studies Reporting on Results of PRP in the Setting of Knee OA[a]

Study	Patients, n	Treatment Groups	Outcome Measures	Results	Conclusions
Lisi, *Clin Rehabil* (2018)[75]	50	Activated PRP vs HA	Modified Shahriaree classification system of knee OA, WOMAC, Lysholm, Tegner, AKSS, Lequesne, VAS_{pain}	48.3% with at least 1 OA grade improvement on repeat MRI for PRP vs 8% in control ($P < .003$). Improved $WOMAC_{total}$, $WOMAC_{ADL}$, AKSS, and Lequesne scores with PRP. No adverse events overall.	Activated PRP reduces articular cartilage damage on MRI as soon as 6 mo after treatment, reduces pain, and improves function.
Cole, *AJSM* (2017)[19]	99	PRP vs HA	$WOMAC_{pain}$, IKDC, VAS_{pain}, Lysholm	No difference in $WOMAC_{pain}$. Improved IKDC in PRP at 24 wk ($P = .013$) and 1 y ($P = .003$). Lower VAS_{pain} at 24 wk ($P = .0096$) and 1 y ($P = .0039$). Mild OA with lower body mass index had improved outcomes.	Results favor PRP over HA for knee OA.
Dai, *Arthroscopy* (2017)[20,b]	1069	PRP vs control	$WOMAC_{pain}$, $WOMAC_{function}$, $WOMAC_{total}$, IKDC, Lequesne	PRP and HA with similar improvements in pain and function at 6 mo. PRP associated with improved pain and function at 1 y, exceeding MCID for $WOMAC_{pain}$ and $WOMAC_{function}$.	Compared with HA and saline, PRP may improve pain relief and function in symptomatic knee OA at 1 y.
Duymus, *KSSTA* (2017)[25]	102	PRP, HA, ozone gas	Radiographic OA progression (Kellgren-Lawrence), WOMAC, VAS_{pain}	Clinical efficacy of PRP and HA similar at 6 mo. PRP with improved VAS and WOMAC scores vs HA at 12 mo ($P < .001$ for each).	PRP more effective than HA and ozone injections at 12 mo for mild to moderate knee OA.
Paterson, *BMC* (2016)[98,c]	23	Photoactivated PRP vs HA	KOOS, KQoL, VAS_{pain}, hopping distance, maximum number of knee bends	No significant between-group differences at 4 or 12 wk.	Photoactivated PRP may provide a safe and effective treatment for knee OA.
Smith, *AJSM* (2016)[124]	30	LP-ACP vs saline	WOMAC	Improved WOMAC at each time point with LP-ACP ($P < .05$). At 1 y, 78% improvement from baseline WOMAC with ACP vs 7% with saline. No adverse events in ACP group.	ACP is safe and improves pain and function in knee OA vs saline control.
Filardo, *AJSM* (2015)[30]	192	PRP vs HA	IKDC, KOOS, EuroQoL, VAS_{pain}, Tegner	Significantly more self-limited postinjection swelling and pain after PRP. No significant intergroup difference at any follow-up time in any clinical measure.	PRP does not provide superior clinical improvement vs HA for knee OA.
Gormeli, *KSSTA* (2017)[53]	162	3 PRP injections, 1 PRP injection, HA	EQ-VAS, IKDC	Significantly better results with early OA treated with 3 PRP injections vs all other groups. No difference between single dose of PRP and HA. No between-group differences in advanced OA.	Multiple PRP injections may achieve better results than HA or a single PRP injection for early knee OA.
Raeissadat, *Clin Med Insights* (2015)[107]	160	PRP vs HA	WOMAC, SF-36	WOMAC (all scales) and SF-36 (all scales) improved in PRP vs HA at 1 y ($P < .001$).	PRP is more effective than HA in reducing symptoms and improving quality of life among select patients with knee OA.
Patel, *AJSM* (2013)[96]	78	1 PRP injection, 2 PRP injections, saline	WOMAC, VAS_{pain}	Significant improvements in WOMAC after 1 or 2 PRP at each time point. No difference in outcomes between 1 and 2 PRP injections. Improved outcomes in early OA.	Single dose of WBC-filtered PRP in concentrations 10× the normal concentration of circulating blood is as effective as 2 injections for early knee OA. Results may deteriorate at 6 mo.
Vaquerizo, *Arthroscopy* (2013)[130]	96	PRGF vs HA	WOMAC, Lequense	Improved WOMAC and Lequense scores at 24 and 48 wk after injection ($P < .001$) in PRGF group vs HA.	PRGF is safe and superior to HA at 24 and 48 wk, with significant clinical improvements in patients with knee OA.
Cerza, *AJSM* (2012)[11]	120	ACP vs HA	WOMAC	Sustained improvement for up to 24 wk with ACP vs HA (WOMAC 65.1 vs 36.5, $P < .001$). No difference in ACP based on severity of arthritis.	Treatment with ACP resulted in significantly better clinical outcomes than HA.

[a]All studies are level 1 randomized controlled trials unless noted otherwise. ACP, autologous conditioned plasma; ADL, Activities of Daily Living; AKSS, American Knee Society Score; EQ-VAS, EuroQol–visual analog scale; HA, hyaluronic acid; IKDC, International Knee Documentation Committee; KOOS, Knee injury and Osteoarthritis Outcome Score; KQoL, Knee Quality of Life Questionnaire; LP, leukocyte poor; MCID, minimal clinically important difference; MRI, magnetic resonance imaging; OA, osteoarthritis; PRGF, platelets rich in growth factors; PRP, platelet-rich plasma; SF-36, 36-Item Short Form Health Survey; VAS, visual analog scale; WBC, white blood cell; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

[b]Meta-analysis (level 1; 10 randomized controlled trials).

[c]Randomized controlled trial (level 2).

TABLE 3
Studies Reporting Outcomes of Cell-Based Therapy for Articular Cartilage Defects of the Knee[a]

Study	Study Design (LOE)	Patients, n	Treatment Groups	Outcome Measures	Results	Conclusions
Gobbi, *KSSTA* (2017)[51]	Prospective cohort (2)	20	BMAC with hyaluronan-based scaffold for ICRS grade 4 chondral lesions vs control	MRI, KOOS, IKDC, VAS, Tegner	Improvements in all KOOS subscales, Tegner, and IKDC subjective in both groups; results were affected by lesion size and number but not concomitant surgery. MRI with complete fill in 80% of treatment group and 71% of control group.	Treatment of cartilage lesions with BMAC and hyaluronan-based scaffold is a viable and effective option affected by lesion size and number but not age.
Oladeji, *AJSM* (2017)[94]	Retrospective cohort (3)	15	Fresh OCA with or without BMAC	Radiographic integration of OCA	BMAC group had higher graft integration scores at all time points ($P < .035$). Nonsmokers had higher graft integration scores at 6 mo ($P = .007$).	Large femoral condylar OCAs treated with BMAC before implantation have improved radiographic integration and may improve OCA bone healing.
Gobbi and Whyte, *AJSM* (2016)[52]	Prospective cohort (2)	50	HA-based scaffold with BMAC vs microfracture	Lysholm, Tegner, IKDC, KOOS	More normal/nearly normal by IKDC score in HA-BMAC ($P < .001$) with sustained improvement only at 5 y. Age >45 y, large lesion size, and multiple lesions not associated with poor outcome with HA-BMAC.	Better clinical outcomes and more durable cartilage repair at medium-term follow-up with HA-based BMAC scaffold vs microfracture.
Krych, *AJSM* (2016)[69]	Retrospective cohort (3)	46	Control scaffold, PRP scaffold, BMAC scaffold	Qualitative and quantitative MRI	Superior cartilage fill in PRP- and BMAC-treated scaffolds vs control ($P < .03$). BMAC group closer to hyaline cartilage than other groups ($P = .01$).	Patients treated with BMAC augmented scaffold had improved cartilage maturation with greater fill, closer to native hyaline cartilage.
Enea, *Knee* (2015)[27]	Case series (4)	9	Microfracture with BMAC collagen membrane cover	IKDC, Lysholm, VAS, histological assessment	Macroscopic cartilage assessment appeared nearly normal at 1-y follow-up; 8 of 9 with significant clinical improvement.	This method appears to be safe and may improve short-term knee function with potential to re-create hyaline-like cartilage.
Gobbi, *Cartilage* (2015)[48]	Prospective cohort (2)	37	MACI and BMAC-derived MSCs in patellofemoral lesions	IKDC, KOOS, VAS, Tegner, MRI, histology	Improved IKDC subjective score with BMAC ($P = .015$), otherwise no between-group differences. No adverse events. MRI with 81% filling in BMAC group.	Both techniques are viable and effective for large patellofemoral chondral lesions.
Gobbi, *AJSM* (2014)[49]	Case series (4)	25	Bone marrow–derived MSCs and second-generation matrix	Radiographs, MRI, VAS, IKDC, KOOS, Lysholm, Marx, Tegner, histology	Improvement in all clinical outcome measures ($P < .001$). Improved outcomes among patients <45 y old with smaller or single lesions. MRI filling in 80%, hyaline-like cartilage on histology. No adverse events.	MSCs are effective in treating large chondral defects in a single-stage procedure.
Enea, *Knee* (2013)[28]	Case series (4)	9	Microfracture covered with resorbable matrix and BMAC	MRI, histology, IKDC, Lysholm, VAS	8 of 9 with clinical improvement. Improvement in all clinical outcomes ($P < .05$). Complete defect filling on MRI. No adverse events.	Technique is safe, improves knee function, and may regenerate hyaline-like cartilage.
Skowronski, *Ortop Traumatol Rehabil* (2013)[123]	Case series (4)	54	BMAC and collagen membrane	KOOS, Lysholm, VAS	Significant clinical improvement in 52 of 54 without any complications.	One-stage reconstruction of large articular cartilage lesions with BMAC is effective at short- to midterm follow-up.
Skowronski and Rutka, *Ortop Traumatol Rehabil* (2013)[122]	Case series (4)	46	Blood-derived stem cells or BMAC with collagen membrane	KOOS, Lysholm, VAS, MRI	Significant improvement in clinical outcomes in 86% of patients. Improved outcomes in patients treated with blood-derived stem cells.	Effective treatment option for full-thickness osteochondral lesions. Inferior outcomes in BMAC group may be attributable to number of cells injected.

(continued)

TABLE 3
(continued)

Study	Study Design (LOE)	Patients, n	Treatment Groups	Outcome Measures	Results	Conclusions
Gobbi, *Cartilage* (2011)[50]	Prospective cohort (2)	15	BMAC and collagen I/III matrix	VAS, IKDC, KOOS, Lysholm, Marx, SF-36, Tegner, MRI, histology	Improvement in all clinical outcomes ($P < .005$). Greater improvement among patients with single and small lesions. Hyaline-like tissue on MRI and histology among all patients analyzed. No adverse events.	Single-stage BMAC and collagen I/III matrix may be a viable option for grade IV knee chondral lesions.

[a]BMAC, bone marrow aspirate concentrate; HA, hyaluronic acid; ICRS, International Cartilage Repair Society; IKDC, International Knee Documentation Committee; KOOS, Knee injury and Osteoarthritis Outcome Score; LOE, level of evidence; MACI, matrix-assisted chondrocyte implantation; Marx, Marx Activity Scale; MRI, magnetic resonance imaging; MSC, mesenchymal stem cell; OCA, osteochondral allograft; PRP, platelet-rich plasma; SF-36, 36-Item Short Form Health Survey; VAS, visual analog scale.

not "structure modifying" (ie, improvement in clinical symptoms but no change in the underlying cartilage pathology).

VARIABILITY OF PREPARATIONS

Premature clinical translation of biologic therapies has resulted in unique challenges, in part because of the use of poorly characterized and often suboptimal treatment preparations.[12,13,71,102] Murray et al[86] recently established an expert consensus on minimum reporting requirements for clinical studies evaluating PRP and MSCs that may critically influence treatment outcome. These consensus requirements should help us to more easily compare outcomes across studies, more accurately reproduce preparations used in published studies, and eventually determine specific, reproducible formulations of biologic therapies for particular clinical indications.

Platelet-Rich Plasma

Peripheral blood is variable in the concentration of growth factors and platelets among patients.[45] Variation in platelet concentrations and other contents is dependent on numerous factors, including harvest site, harvest volume, patient demographics (eg, age and sex), concomitant medical comorbidities, medications, nutritional factors, and recent activity level.[136,138,139] Furthermore, PRP preparation protocols include multiple stages that may each vary, including sample volume, type of anticoagulation, and centrifuge device characteristics and settings (eg, *g* forces, number of spin cycles, duration of spin cycles, and method of separation among serum, cell, and platelet fractions).[12] For example, among clinical studies reporting on the use of PRP to augment rotator cuff repair, the PRP preparation process and application method varied widely,[8,14,109,113,114,134] with studies administering PRP via a 1-time injection during repair,[12,14,109,114] by suturing PRP (as platelet-rich fibrin matrices) into a scaffold between tendon and bony footprint,[8,109,113,134] and as a suture coated in PRP,[65] thereby making comparison among studies difficult.

Multiple studies attempted to determine the optimal concentration of platelets for musculoskeletal healing.[35,135,140] Although the efficacy of PRP likely depends not only on platelet content/concentration but also on leukocyte concentration,[112] no randomized or prospective studies have been performed comparing outcomes between LR-PRP and LP-PRP.[68] In an effort to determine the optimal leukocyte concentration in PRP for knee OA, 1 meta-analysis of RCTs comparing LR-PRP and LP-PRP revealed that 3 RCTs with LP-PRP demonstrated superior outcomes versus HA, while only 1 RCT with LR-PRP reported superior outcomes over HA.[112] Fitzpatrick et al[34] performed a study of 4 commercially available PRP kits, reporting variation in blood components, including platelets, red blood cells, and leukocytes, as well as pH and glucose in the preparations. The authors concluded that the lack of standardization of PRP preparations for clinical use has likely contributed to the varying clinical efficacy of PRP.[34] The authors also stated that because different preparations of PRP are likely to be important in the management of different pathologies, all commercial kits should be validated for cell and PRP types.[22,34]

The current literature evaluating PRP fails to include sufficient experimental detail or report the basic composition of PRP formulations.[12,87] Chahla et al[12] found that only 10% of clinical trials reporting on the use of PRP in orthopaedic conditions provided comprehensive reporting, including a clear description of the preparation protocol that could be used by other investigators to repeat the methods, and only 16% provided quantitative data on the composition of the final PRP product. Reporting of processing steps should be standardized to allow for comparison and outcome assessment among studies and reproduction of results.[12]

Bone Marrow Aspirate Concentrate

The viability and efficacy of BMAC and other cell therapies are also affected by harvest site, cell concentration, and donor characteristics, including age, sex, and comorbidities.[17,78,99]

Pocket Guide for the Clinical Use of Biologic Therapies in Sports Medicine Conditions

Platelet-rich Plasma

Common clinical indications#: Rotator cuff tear, epicondylitis, muscle injury, tendinopathy/tendon injury, ACL, OA, osteochondral lesion

Variables for Consideration[1]

Starting volume (whole blood sample)

Type of anticoagulant used

- E.g. Acid citrate dextrose solution A (ACD-A), calcium citrate, citric acid, citrate phosphate dextrose, sodium citrate

Method of separation and processing machine used

- Centrifugation (*most common*)
- # of spin cycles (generally 1-2 spins)
- Duration of spin cycle(s)
- Spin rate (RPM) and/or G-forces
  - Spin 1: Most commonly ≈3,200 RPM x 15 min
  - Spin 2: Most commonly ≈3,200 RPM x 10 min

Desired concentration of platelets for clinical indication

- **Leukocyte-rich (LR-PRP) vs Leukocyte-poor (LP-PRP)**

Platelet-activation method

- E.g. $CaCl_2$, thrombin, dry needling, calcium gluconate, tissue factor

Use of buffer ($NaHCO_3$)

Final composition

- **Total cell count: Leukocytes, erythrocytes, platelets**
- **Platelet count: Most commonly ≈1,000 x 10^3 (5-fold increase)**

Final volume: Most commonly ≈3-5 mL

Time from preparation to use: Most commonly ≈ 1 hour

Bone marrow aspirate concentrate (BMAC)

Common clinical indications#: Osteochondral lesion, OA, rotator cuff tear, ACL tear, epicondylitis, tendinopathy

Variables for Consideration[2]

Collection site

- **Most commonly iliac crest (highest yield of progenitors[3])**
- Number of sites aspirated: Most commonly ≈ 4 sites (≈5.5 mL/site)

Starting volume of bone marrow: Most commonly ≈60 mL

Final volume of BMAC injected: Most commonly ≈6 mL

Anticoagulant used (Heparin, ACD-A)

Initial/Final nucleated cell count

- Initial: Most commonly ≈4.7x10^6 cells/mL
- **Final: Most commonly ≈24.9 x10^6 cells/mL**
- **BMAC colony count: ≈1,134 CFU/mL**

Processing machine and settings

- # of spin cycles
- Duration of spin cycles
- Spin rate (RPM) and/or G-forces
  - Most commonly single spin at 3,200 RPM x 15 mins

Patient-specific factors may impact yield

Age: Higher in younger patients[4]
Sex: Higher in males[4]
Medical comorbidities
Medications
Nutritional status
Activity level: Exercise may Increase platelet concentration in PRP [5]
Harvest site and sample volume

#Refer to text for details on clinical indications and overview of clinical outcomes
[1]Chahla et al[11]
[2]Piuzzui et al[102]
[3]Hyer et al[61]
[4]Xiong et al[139]
[5]Anz et al[5]

Figure 2. Pocket guide for the clinical use of biologic therapies in sports medicine conditions. ACL, anterior cruciate ligament; OA, osteoarthritis; PRP, platelet-rich plasma; RPM, revolutions per minute.

Because preparations of BMAC vary per the anatomic location harvested, harvesting strategy, and unique processing method, BMAC preparations vary widely in the concentration of progenitors and non-MSC cell types, which can influence MSC behavior and growth factor content.[56,89,90] Currently, there remains a lack of understanding on the optimal preparation, anatomic source, delivery method, timing, and dosing of BMAC preparations.[86] There are >14 commercially available protocols with unique compositions and characteristics, and few studies have investigated and compared the variability among preparations from these different point-of-care systems.[47,89] Piuzzi et al[102] performed a systematic review of literature investigating the use of BMAC in musculoskeletal conditions, and no study provided a clear description of the preparation protocol that could be used by other investigators to reproduce the methods.

There is a critical need for accurate and standardized reporting of preparation protocols and the resulting cell content for all biologic preparations in the peer-reviewed literature. Recent studies clearly defined this issue, established consensus reporting criteria, and encouraged investigators of future studies to adhere to these important standards.[12,86,89,102] Figure 2 provides an overview of the 2 most commonly used biologic therapies, PRP and BMAC, with important variables for consideration with use of these therapies.

CONCLUSION

The clinical use of biologic therapies has increased in the United States in recent years even though their clinical efficacy and cost-effectiveness have not been well established owing to an insufficient number of well-designed clinical trials and a lack of standardization in biologic preparations and outcomes across studies.[70,80,100,104,105,142] Clinicians utilizing or considering the use of biologic therapies must be able to answer basic questions about treatment costs, clinical efficacy, and associated risks.[93,137] While current evidence suggests that there likely is a role for biologic therapies in the treatment of musculoskeletal conditions, these treatments are still within the "proof of concept" stage, and we must recognize and close

the gap between what is currently known and what is marketed to patients.[104,105] Beyond the current evidence, clinicians must understand what is currently marketed to patients and be able to delineate discrepancies between current clinical evidence and marketing claims.[104] Direct-to-consumer advertising of biologic therapies by clinics and commercial entities portrays an almost-exclusively positive tone, without providing a balanced view of the risks, benefits, and limitations of these treatments.[108] However, it is imperative that as clinicians we are aware of the discrepancies between the scientific evidence available and the marketing claims to appropriately counsel our patients.

Currently, there is a lack of consensus on the optimal preparation, source, delivery method, and dosing of biologic therapies in sports medicine applications, and this is exacerbated by a lack of sufficient experimental detail in most published studies.[12,86,89] Ultimately, we need to identify the "biologic target" for different tissues and diverse pathologies. The ideal therapy depends on the goal of treatment. Do we want to accelerate cell proliferation? Stimulate matrix synthesis? Inhibit inflammatory mediators and matrix-degrading enzymes? Stimulate angiogenesis? Identification of the treatment goal will allow eventual matching of the optimal cell or PRP formulation to the specific tissue and pathology. Clearly, one size does not fit all. Future research must not only adhere to better standards of reporting but also better integrate researchers, industry, and regulatory bodies such as the FDA in an effort to more efficiently and effectively develop safe and improved therapies for patients.[71]

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